

— ML Performance (2026-05-01 s/d 2026-06-13)

Metric	Value
Evaluation Generated At (UTC)	2026-04-28T02:09:13.062081+00:00
=== 1. MODEL PERFORMANCE SUMMARY ===	
--- AGENUSA ---	
Active Model Version	agenusa_v20260608161639411699
Model Created At	2026-06-08 16:16:37
Dataset Size	5000
Anomaly Count (default)	400
Anomaly Rate	8.0%
Contamination Rate	0.08
Review Score Threshold	0.001367
High-Risk Score Threshold	-0.005265
Predictions at Review Boundary	500
Numeric Features Count	17
Categorical Features Count	3
Numeric Features	AMOUNT, PROCESSING_CODE, RESPONSE_CODE, MTI, TX_HOUR, TX_DAYOFWEEK, IS_NIGHT_TX, GAP_MINUTES, TERMINAL_SWITCH_FAST, AVG_AMOUNT_5, AMOUNT_OVER_AVG_RATIO, IS_DECLINED, IS_BRUTE_PATTERN, IS_MONEY_MULE_DEST, IS_HIGH_AMOUNT_PATTERN, MIDNIGHT_AMOUNT_SPIKE, RAPID_RETRY_DECLINED
Categorical Features	TERMINAL_ID, MERCHANT_ID, DEST_ACCOUNT_NUMBER

--- NUSABILL ---	
Active Model Version	nusabill_v20260605030000541329
Model Created At	2026-06-05 03:00:00
Dataset Size	5000
Anomaly Count (default)	500
Anomaly Rate	10.0%
Contamination Rate	0.1
Review Score Threshold	-0.0
High-Risk Score Threshold	-0.065467
Predictions at Review Boundary	500
Numeric Features Count	12
Categorical Features Count	2
Numeric Features	BILL_AMOUNT, PAYMENT_AMOUNT, REFUND_FLAG, PAYMENT_DELAY_DAYS, PAYMENT_TO_BILL_RATIO, UNDERPAY_FLAG, HIGH_SPIKE_FLAG, CHANNEL_API_FLAG, PAYMENT_GAP_MINUTES, BURST_FLAG, CHANNEL_SWITCH_TO_API, EARLY_PAYMENT_ANOMALY
Categorical Features	CHANNEL, BILL_STATUS
=== 2. RETRAIN HISTORY ===	Periode: 2026-05-01 s/d 2026-06-13

Metric	Value
Total Retrain Executions	20
Successful Retrains	20
Retrain Success Rate (%)	100.0
Total Records Trained (cumulative)	15000
Total Feedback Records Used (cumulative)	0
Total Anomalies Found (cumulative)	5000
Total New Patterns Discovered (cumulative)	1
Trigger Type Breakdown	
- manual_upload	20
Recent Retrain Executions (max 10)	
Execution Time Trigger Status Records Feedback Used New Patterns	
2026-06-08 16:16:39 manual_upload SUCCESS	records=-, feedback=-, patterns=0
2026-06-08 15:24:48 manual_upload SUCCESS	records=-, feedback=-, patterns=0
2026-06-08 15:03:54 manual_upload SUCCESS	records=-, feedback=-, patterns=0
2026-05-11 14:37:05 manual_upload SUCCESS	records=-, feedback=-, patterns=0
2026-05-11 14:36:56 manual_upload SUCCESS	records=-, feedback=-, patterns=0
2026-05-11 14:36:51 manual_upload SUCCESS	records=-, feedback=-, patterns=0
2026-05-11 14:36:44 manual_upload SUCCESS	records=-, feedback=-, patterns=0
2026-05-11 14:36:36 manual_upload SUCCESS	records=-, feedback=-, patterns=0
2026-05-11 14:36:18 manual_upload SUCCESS	records=-, feedback=-, patterns=0
2026-05-11 14:36:10 manual_upload SUCCESS	records=-, feedback=-, patterns=0
=== 3. PATTERN DISCOVERY ===	
Top 10 Patterns Category Risk Score Hit Count	
Velocity Fraud Suspect	category=VELOCITY, risk_score=80, hits=150
High Amount Spam Nusa	category=AMOUNT, risk_score=70, hits=80
[Auto Cluster] Potential Fan-In Card Collusion	category=NETWORK, risk_score=60, hits=5
Auto High Amount AGENUSA (5000000+)	category=AMOUNT, risk_score=50, hits=0
AI Discovery: Burst Payment	category=BURST_ATTACK, risk_score=80, hits=0
BURST_AMOUNT	category=BURST, risk_score=50, hits=0
AI Discovery: Brute Force PIN	category=BRUTE_FORCE, risk_score=85, hits=0
AI Discovery: Rapid Retry Declined	category=VELOCITY, risk_score=80, hits=0
af	category=NETWORK_FAN_IN, risk_score=50, hits=0
Decline Velocity	category=DECLINE_VELOCITY, risk_score=85, hits=0
Total Patterns in System	17
Active Patterns	14
=== 4. FRAUD DETECTION EFFECTIVENESS ===	Periode: 2026-05-01 s/d 2026-06-13

Metric	Value
--- AGENUSA ---	
Total Transactions Processed	329
Actual Fraud Count	52
Actual Fraud Rate (%)	15.81
Model Anomaly Rate (%)	8.0
Anomaly Rate vs Actual Fraud Rate Gap	-7.81 pp (model kurang sensitif)
--- NUSABILL ---	
Total Transactions Processed	211
Actual Fraud Count	39
Actual Fraud Rate (%)	18.48
Model Anomaly Rate (%)	10.0
Anomaly Rate vs Actual Fraud Rate Gap	-8.48 pp (model kurang sensitif)
Total New Patterns Generated (all retrains, periode ini)	1
=== CATATAN ===	
Akurasi Model	Tidak tersedia — dataset training bersifat unlabeled (tanpa kolom IS_FRAUD), sehingga metrik Precision, Recall, dan F1-Score tidak dapat dihitung.
Metode Evaluasi	Berbasis threshold boundary dari hasil training (Isolation Forest), bukan validasi terhadap ground truth.
Anomaly Rate vs Fraud Rate	Perbandingan ini bersifat indikatif — anomaly rate adalah parameter model (contamination), sedangkan fraud rate adalah hasil keputusan final (rule engine + pattern + ML + review analisis).